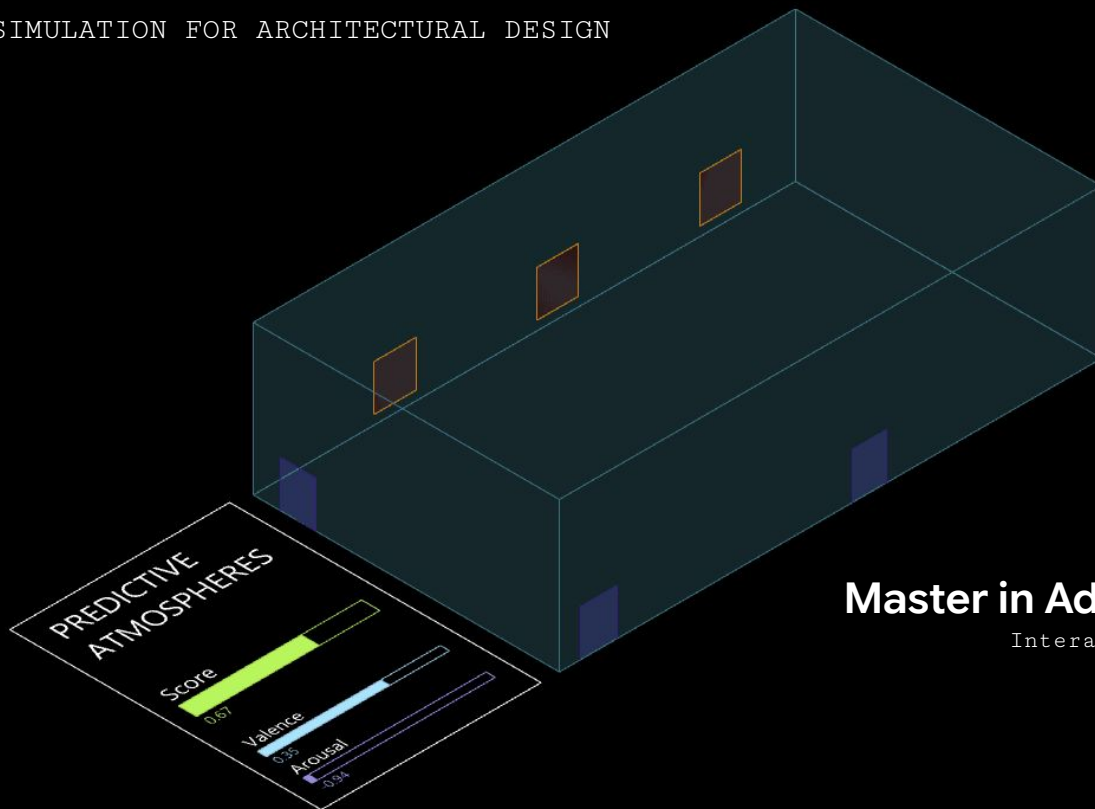


PREDICTIVE ATMOSPHERES

EMOTIONAL SIMULATION FOR ARCHITECTURAL DESIGN



Master in Advanced Architecture 02

Interactive Architecture Thesis Cluster

MANIDEEP MAMIDALA

Thesis Advisors :

Firas Safieddine

Francisco Ramazzini Estivallet

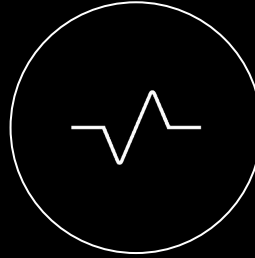
The Static Environment Problem

Current Situation



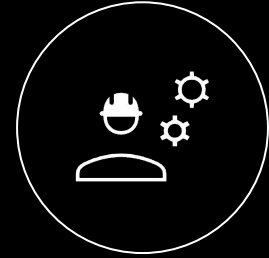
90% of Life Indoors

The vast majority of time is spent in built environments.



Modeling Softwares

Current modeling software and feedback systems fail to account for the physiological and psychological states of the occupants.



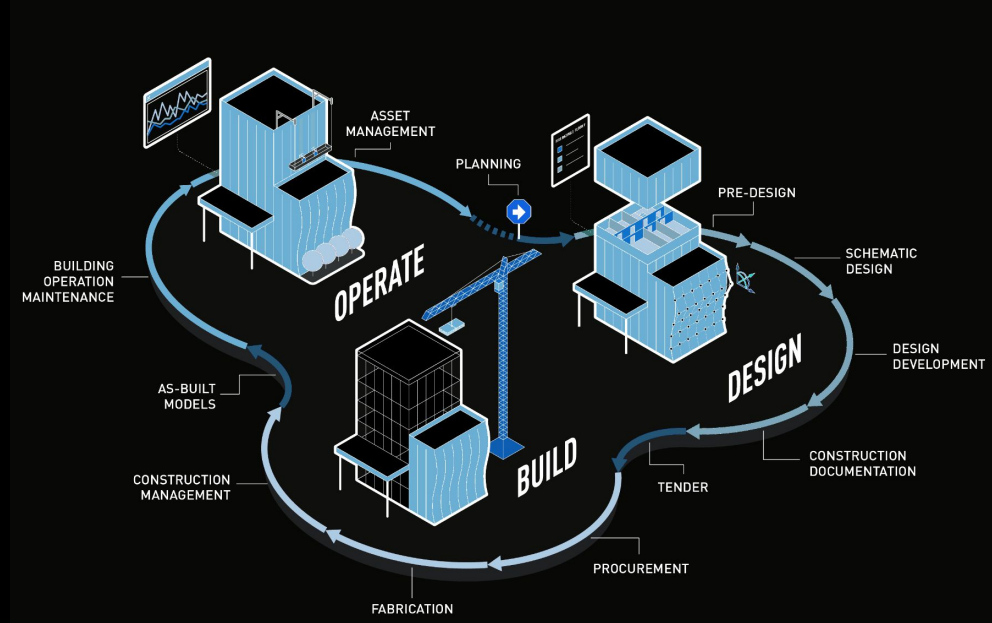
High Barrier to Entry

Evaluating these metrics requires designers to purchase specialized equipment, master complex software, and correctly interpret scientific data.

BIM & WELL

Current Situation

Building Information Modelling (BIM)



WELL Certification



THE WELL BUILDING STANDARD™

SEVEN CONCEPTS FOR HEALTHIER BUILDINGS



AIR



WATER



NOURISHMENT



LIGHT



FITNESS



COMFORT



MIND

2017 © INTERNATIONAL WELL BUILDING INSTITUTE PBC

Research Question

How can we develop **predictive models that infer emotional states from spatial and human metrics**, enabling a **new generation of emotionally aware BIM** tools that critique and extend frameworks like WELL certification?

Time Field

**Publications/
Literature**

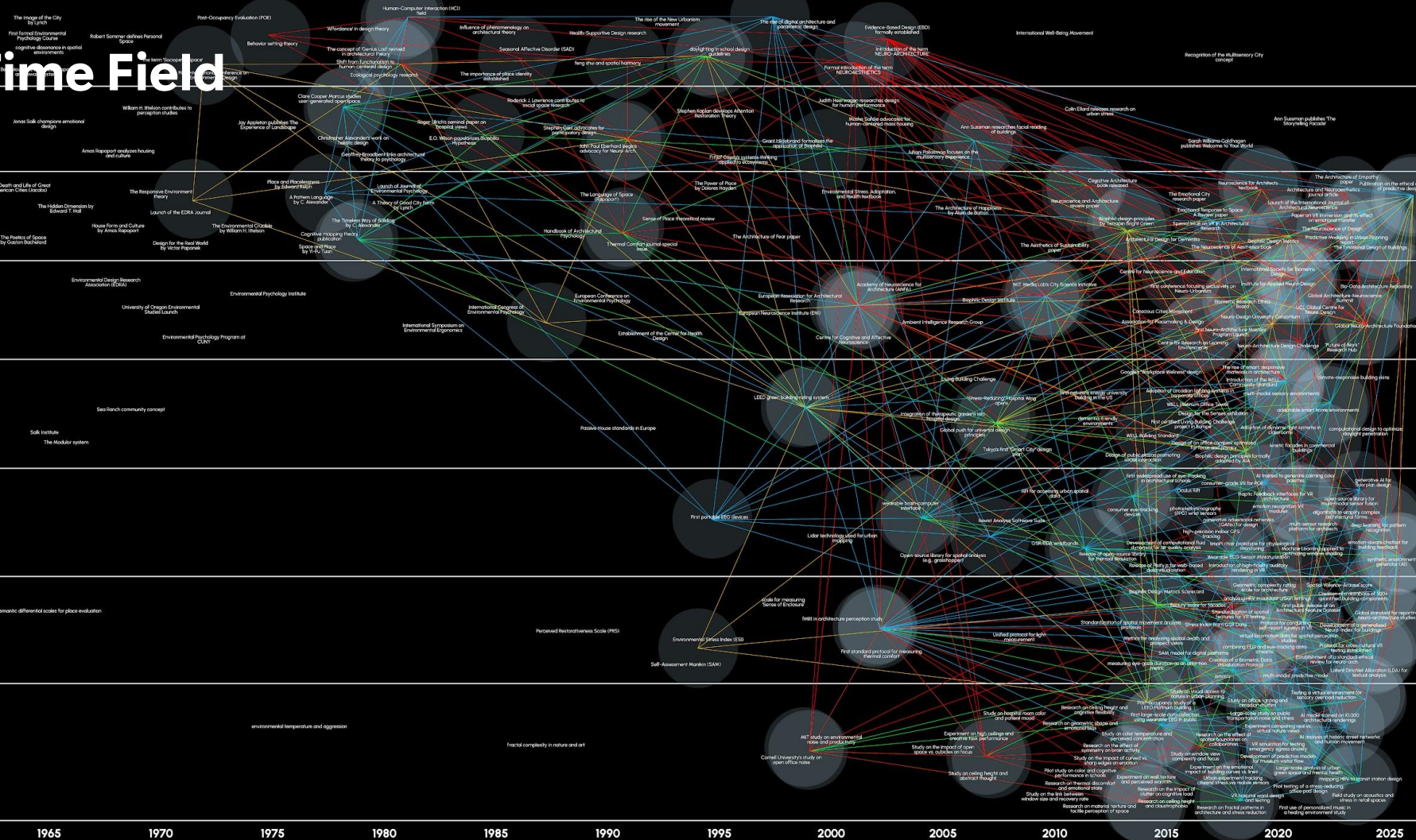
Institutions/ organizations

Architecture/ Design

Technologies/ Tools

Database/
Methods

Research/ Studies



State of the Art

Methodology

The Privacy Paradox in Residential Design

Collective Spaces (Living Rooms)

- Boost psychological dominance and pleasure.
- Lower physiological stress.

Private Spaces (Bedrooms & Bathrooms)

- Trigger significantly higher physiological arousal.
- Caused by high cognitive load and reduced architectural privacy (e.g., floor-to-ceiling glass).



Figure 7 | Graphical representation of the mean values of physiological responses (Frontal-Beta, LF/HF and Phasic) during the visual exploration of the dwelling, in each of the evaluated rooms.

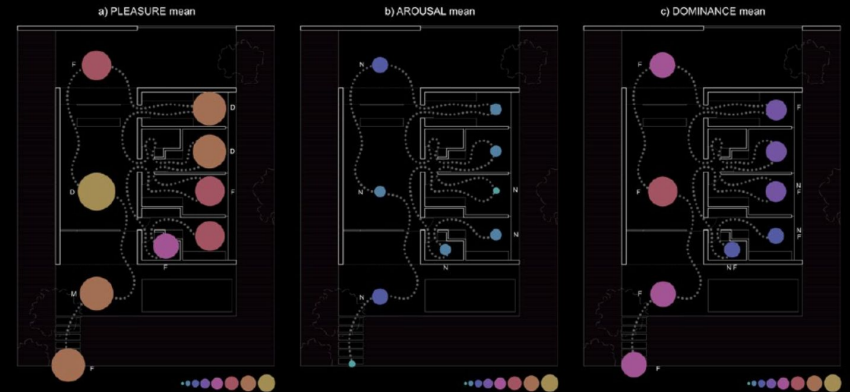


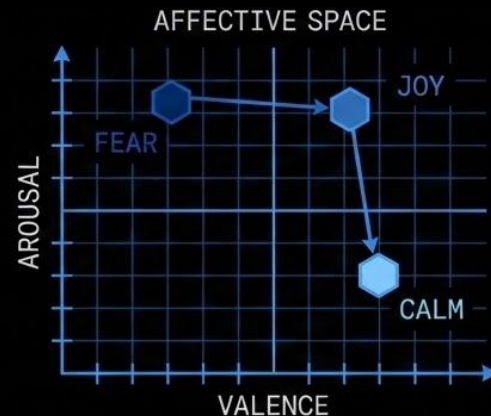
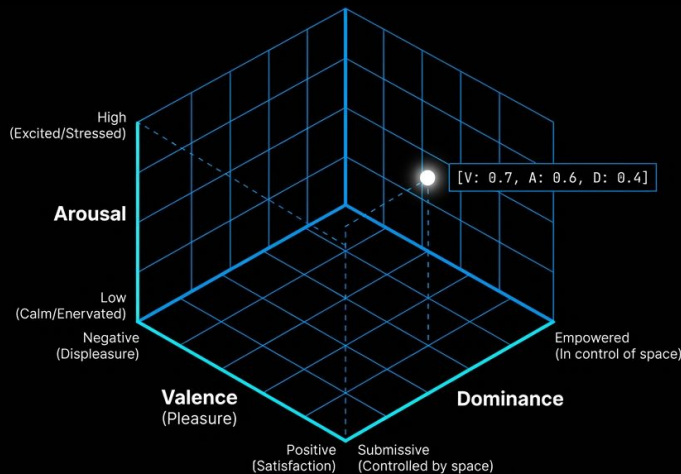
Figure 6 | Graphical representation of the mean values of psychological responses (arousal, pleasure and dominance) during the visual exploration of the dwelling, in each of the evaluated rooms.

State of the Art

Methodology

The V.A.D. Model : We Navigate Emotions Like We Navigate Rooms

- Hippocampal-prefrontal circuits map physical space directly to emotional concepts.
- Emotions are mapped on 2D affective space of valence and arousal.
- The brain tracks trajectories of pleasantness and stress in a building just like physical GPS coordinates.

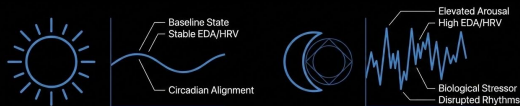


State of the Art

Findings

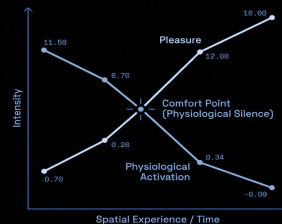
The Temporal Dimension of Space

- Spatial perception is governed by human circadian rhythms.
- Nighttime exposure to residential spaces triggers significantly increased physiological arousal (elevated EDA and HRV).



The Core Paradox of Spatial Comfort

- Spaces eliciting the highest psychological pleasure generate the lowest physiological activation.
- True environmental comfort equates to physiological silence.



The Neuroscience of Space

Curvilinear Forms

Promotes approach behaviors, psychological safety, and calm.



Rectilinear & Sharp Forms

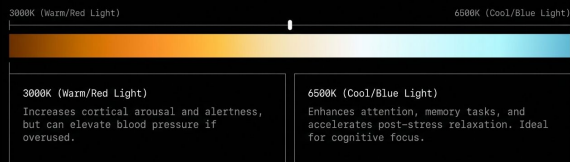
Heightens sympathetic arousal. Interpreted by the brain as environmental stressors.



Sculpting Mood with Photons

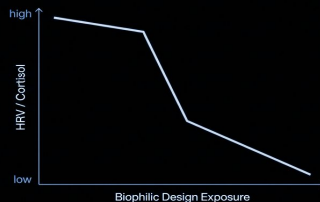
Increases cortical arousal and alertness but can elevate blood pressure if overused.

Enhances attention, memory tasks and accelerates post-stress relaxation.



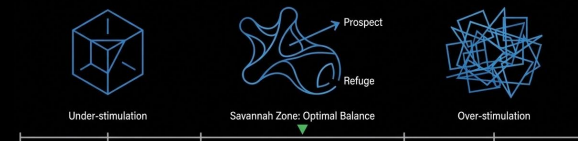
The Biological Craving for Nature

- The "Nature Effect" is a measurable physiological response, not just an aesthetic preference.
- Natural spaces overwhelmingly correlate with happy memories and post-stress recovery destinations.



Complexity, Prospect, and Refuge

- Visual Complexity: The brain prefers moderate complexity.
- Prospect: The ability to have wide visual control.
- Refuge: The places to find psychological shelter.



Hypothesis

If machine learning models are trained on linked datasets of spatial and human metrics, then architecture can integrate emotional inference as a measurable design layer, making BIM integrate emotional metrics and predict human scores.

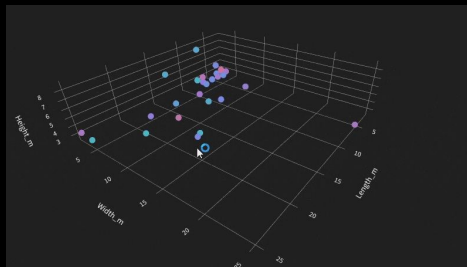
Expected Outcomes & Vision

Key Deliverables



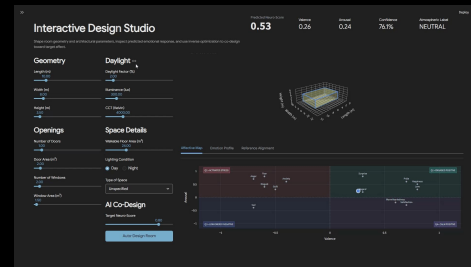
Quantified Human & Spatial Metrics

A categorised, structured database of architectural features (geometry, acoustics, lighting) linked to psychological metrics.



Predictive AI Model

A machine learning pipeline that takes spatial features as input and outputs a predicted emotional state (e.g., valence/arousal score).



Prototype Tool/Platform

A proof-of-concept interface allowing architects to test design parameters and instantly receive emotional impact scores.

THE EXPERIMENTS

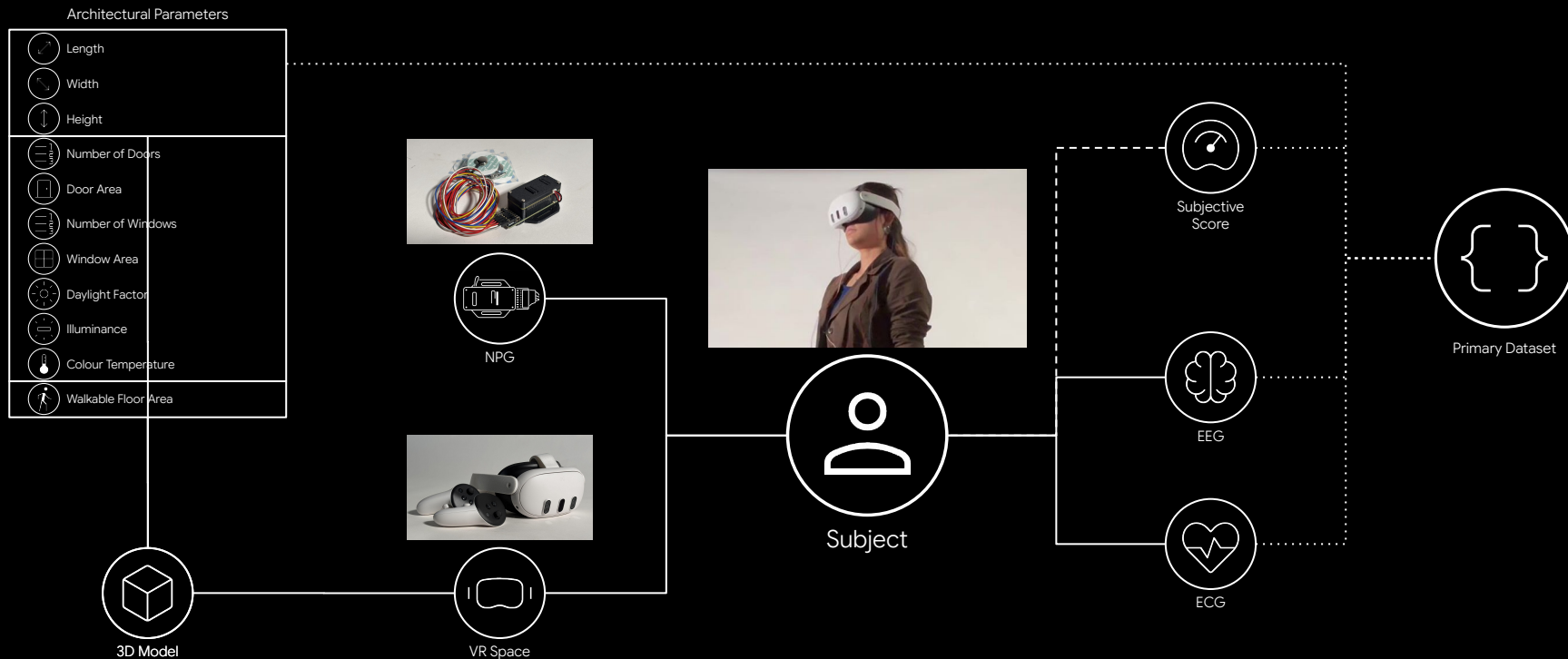
Experiment Design

Methodology

Spatial Design & Metrics

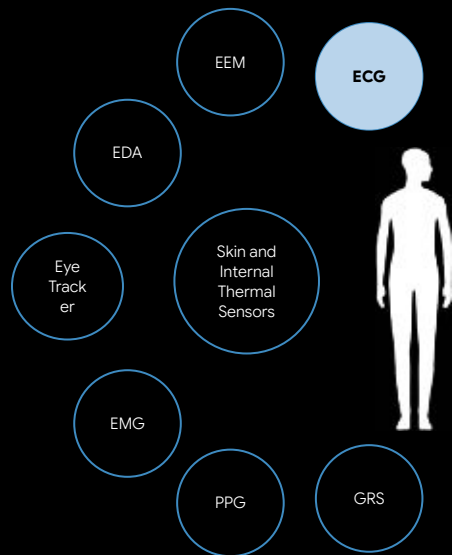
Extracting Human Metrics

Preparing Dataset

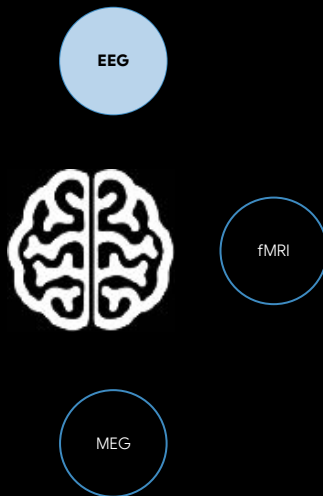


Human Metrics

All possible metrics

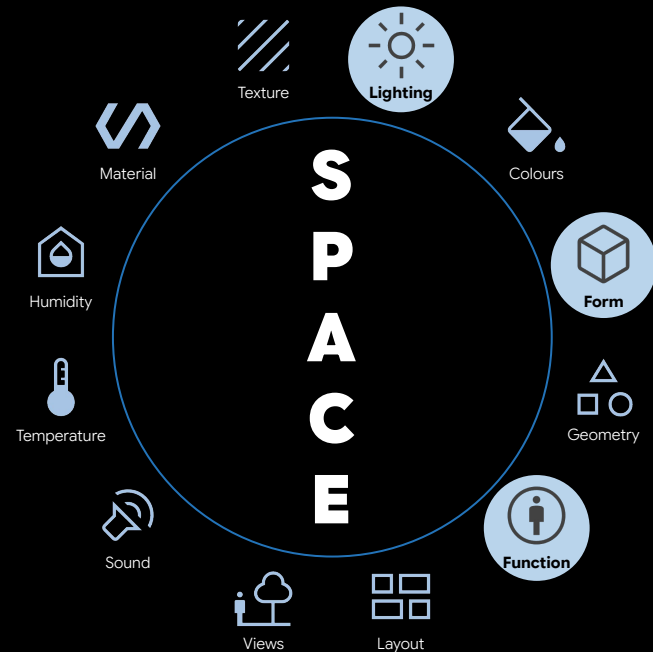


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Spatial Metrics

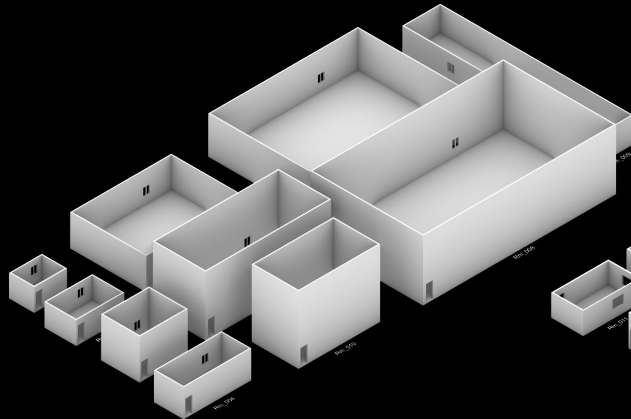
All possible metrics



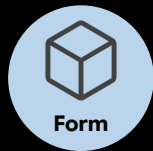
Experiments

Architectural Details

Experiment - 01



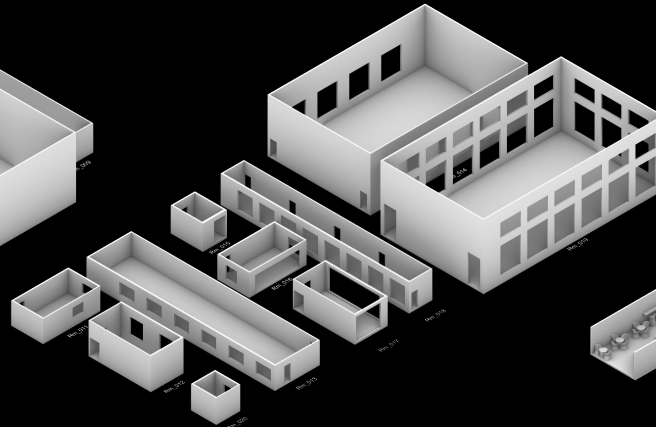
10 Spaces



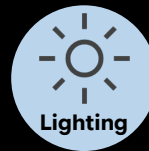
Form

Length
Width
Height

Experiment - 02



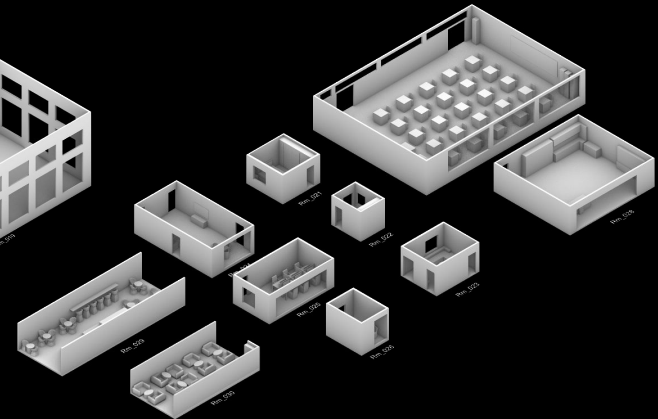
10 Spaces



Lighting

Doors & Windows
Daylight Factor
Illuminance
Colour Temperature

Experiment - 03



10 Spaces



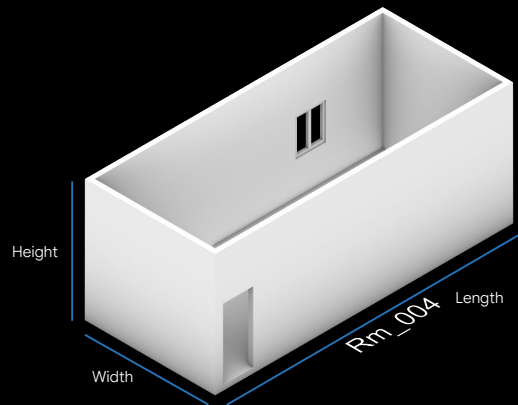
Function

Function of the Space
Walkable Floor Area

Experiments

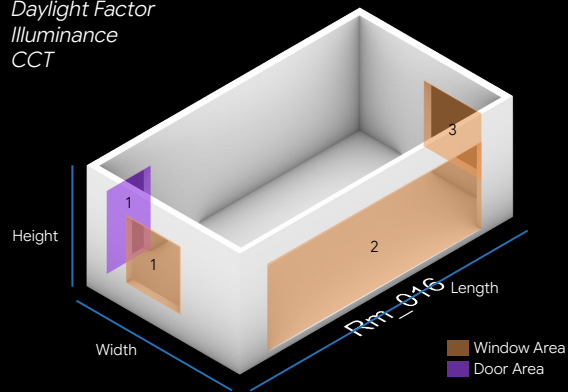
Architectural Details

Experiment - 01



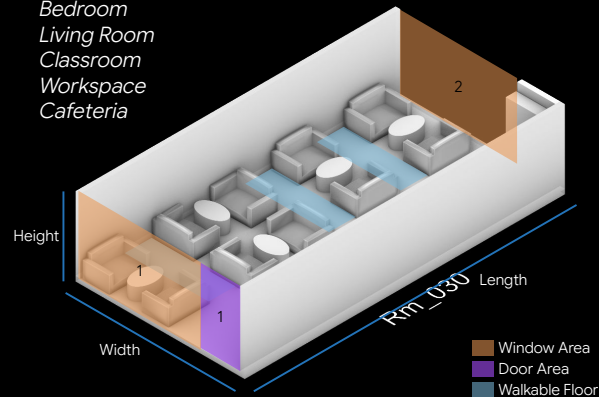
Experiment - 02

Daylight Factor
Illuminance
CCT



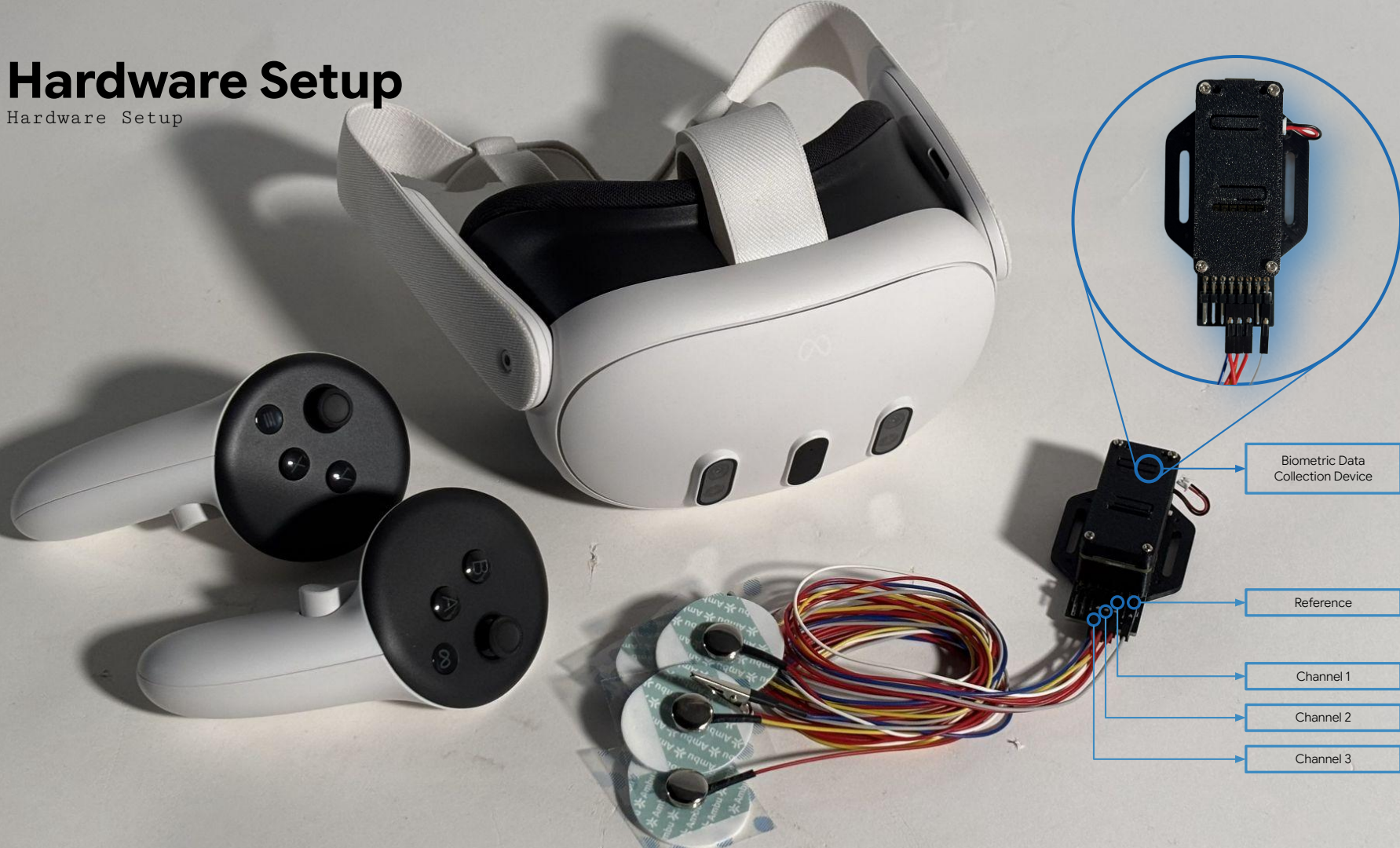
Experiment - 03

Bedroom
Living Room
Classroom
Workspace
Cafeteria



Hardware Setup

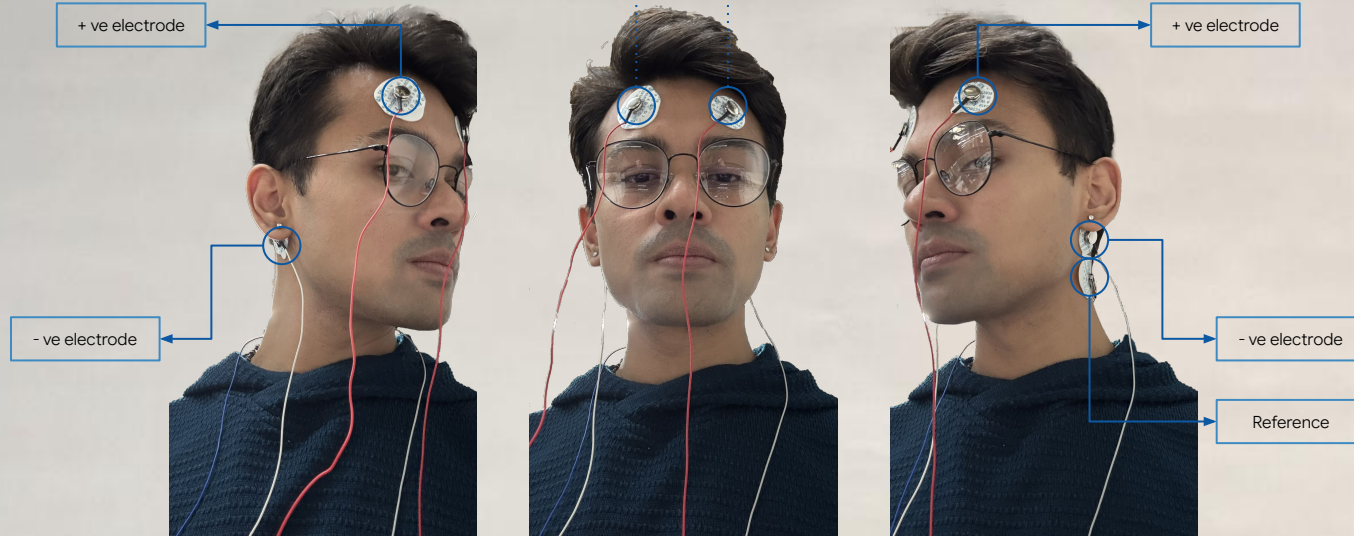
Hardware Setup



Experiment Documentation

Physiological Data Collection

Channel 1 ← → Channel 2

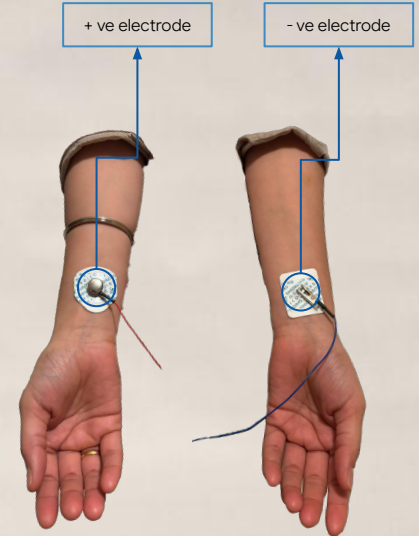


Channel 1
Frontal Right Lobe
EEG

Front

Channel 2
Frontal Left Lobe
EEG

Channel 3



Channel 3
ECG

Experiment Documentation

The Context



Experiment Documentation

The Subject



Experiment Documentation

The Subject



Chords

Channel 1 - Right brain

Channel 2 - Left brain

Channel 3 - Heart

Disconnect



Experiment Documentation

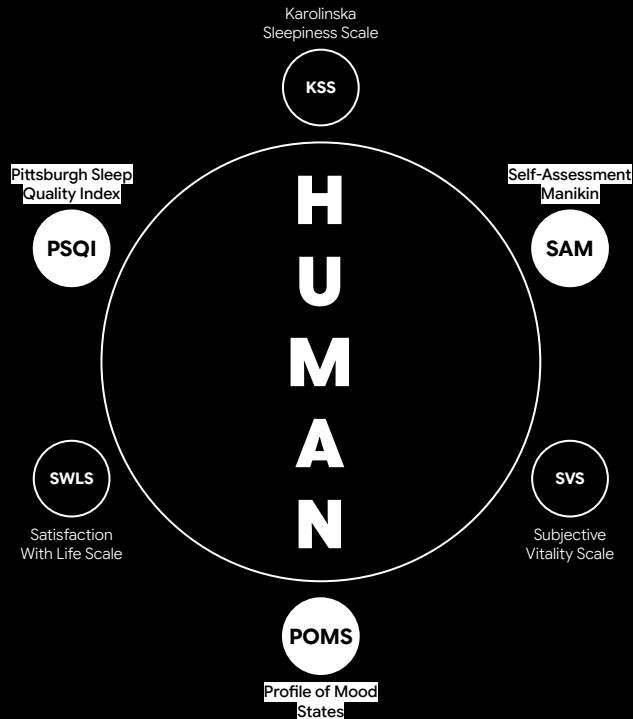
The Subject



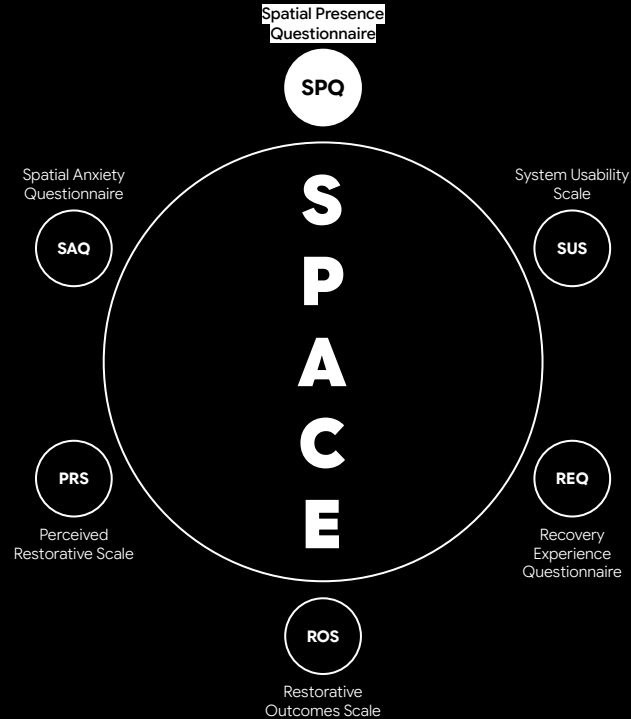
Subjective Data

Data Collected from the Subject

Human Metrics



Spatial Metrics



THE DATABASE

Correlated Datasets of Human + Spatial Metrics

Datasets created from the experiments

Core Dataset of an Experiment

	A	B	C	D	E	F	G
	Subject_ID	Room_ID	Time spent by the subject in the room (seconds)	EEG_Filename	Valence Score by Subject	Arousal Score by Subject	Sleep Hours
1	Subject_A	Rm_021	51	Subj_A-20260420-170123.csv	0.85	-0.2	8
2	Subj_A	Rm_022	60	Subj_A-20260420-171246.csv	-0.6	1	8
3	Subj_A	Rm_023	60	Subj_A-20260420-175231.csv	0.7	0.6	8
4	Subj_A	Rm_024	60	Subj_A-20260420-180836.csv	0.9	-0.8	8
5	Subj_A	Rm_025	60	Subj_A-20260420-181440.csv	0.8	-0.4	8
6	Subj_A	Rm_026	60	Subj_A-20260420-181744.csv	-0.2	0.2	8
7	Subj_A	Rm_027	60	Subj_A-20260420-182152.csv	0.8	0.6	8
8	Subj_A	Rm_028	60	Subj_A-20260420-182546.csv	-0.1	-0.2	8
9	Subj_A	Rm_029	60	Subj_A-20260420-182829.csv	0.82	0.6	8
10	Subj_A	Rm_030	60	Subj_A-20260420-183112.csv	0.94	-0.6	8
11	Subj_M	Rm_030	60	Subj_M-20260420-184241.csv	0.75	-0.2	8.5
12	Subj_M	Rm_029	60	Subj_M-20260420-184540.csv	0.4	0.34	8.5
13	Subj_M	Rm_028	60	Subj_M-20260420-184846.csv	-0.1	0.52	8.5
14	Subj_M	Rm_027	60	Subj_M-20260420-185101.csv	0.4	-0.1	8.5
15	Subj_M	Rm_026	60	Subj_M-20260420-185353.csv	0.36	-0.1	8.5
16	Subj_M	Rm_025	60	Subj_M-20260420-185626.csv	0.8	0.2	8.5
17	Subj_M	Rm_024	60	Subj_M-20260420-185929.csv	0.9	-0.74	8.5
18	Subj_M	Rm_023	60	Subj_M-20260420-190210.csv	0.65	-0.1	8.5
19	Subj_M	Rm_022	60	Subj_M-20260420-190438.csv	-0.8	0.8	8.5
20	Subj_M	Rm_021	60	Subj_M-20260420-190648.csv	0.8	-0.6	8.5
21	Subj_D	Rm_021	60	Subj_D-20260502-163708.csv	0.7	-1	7.5
22	Subj_D	Rm_022	60	Subj_D-20260502-164031.csv	-0.8	-0.2	7.5
23	Subj_D	Rm_023	60	Subj_D-20260502-164331.csv	0.6	-0.6	7.5
24	Subj_D	Rm_024	60	Subj_D-20260502-164715.csv	0.9	-1	7.5
25	Subj_D	Rm_025	60	Subj_D-20260502-164946.csv	0.5	0.2	7.5
26	Subj_D	Rm_026	60	Subj_D-20260502-165445.csv	-0.2	0.4	7.5
27	Subj_D	Rm_027	60	Subj_D-20260502-165721.csv	0.6	-0.2	7.5
28	Subj_D	Rm_028	60	Subj_D-20260502-170024.csv	0.6	0.7	7.5
29	Subj_D	Rm_029	60	Subj_D-20260502-170255.csv	0.8	-0.4	7.5
30	Subj_D	Rm_030	60	Subj_D-20260502-170531.csv	-0.8	0.6	7.5
31	Subj_B	Rm_030	60	Subj_B-20260502-182434.csv	0.5	-0.8	7
32	Subj_B	Rm_029	60	Subj_B-20260502-182744.csv	0.8	-0.85	7
33	Subj_B	Rm_028	60	Subj_B-20260502-183005.csv	0.65	-0.8	7
34	Subj_B	Rm_027	60	Subj_B-20260502-183220.csv	0.65	0.4	7
35	Subj_B	Rm_026	60	Subj_B-20260502-183441.csv	-0.3	-0.2	7
36	Subj_B	Rm_025	60	Subj_B-20260502-183742.csv	0.25	-0.45	7
37	Subj_B	Rm_024	60	Subj_B-20260502-184007.csv	0.5	-0.6	7
38	Subj_B	Rm_023	60	Subj_B-20260502-184247.csv	0.4	-0.3	7

experiment_03_bic

Standard Spatial Data of an Experiment

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
																		Walkable	
		Length	Width	Height	Wall Area (sq.meter)	Floor Area (sq.meter)	Number of Door	Door Area (sq.meter)	Door to Wall Ratio (%)	Number of Windows	Window Area (sq.meter)	Window to Wall Ratio (%)	Daylight Factor (%)	Illuminance (lux)	Correlated Color Temperature (Kelvin)	Day or Night	Walkable Floor Area (sq.meter)	Floor Area to Total Floor Area	Type of Space
1	Room_ID																		
2	Rm_021	3.51	3.87	2.59	38.20	13.57	1.00	1.80	4.71	1.00	2.14	5.60	1.44	350.00	5500 Day		8.90	65.60	Bedroom
3	Rm_022	2.52	2.60	2.80	28.67	6.55	1.00	1.69	5.89	2.00	3.43	11.96	3.93	450.00	5500 Day		4.13	63.03	Bedroom
4	Rm_023	4.77	3.91	2.79	48.35	18.65	3.00	6.58	13.61	1.00	2.08	4.30	1.04	400.00	5500 Day		12.80	68.63	Living Room
5	Rm_024	4.73	8.92	2.56	69.89	42.19	4.00	8.10	11.59	1.00	7.85	11.23	2.34	350.00	5500 Day		33.15	78.57	Living Room
6	Rm_025	7.33	3.99	2.59	58.64	29.25	1.00	1.80	3.07	2.00	2.43	4.14	6.38	900.00	5500 Day		12.45	42.57	Workplace
7	Rm_026	2.87	3.86	2.94	39.57	11.08	1.00	2.28	5.76	1.00	3.84	9.70	3.05	350.00	5500 Day		4.38	39.54	Workplace
8	Rm_027	18.50	13.16	3.36	212.76	243.46	2.00	9.48	4.46	13.00	43.90	20.63	2.46	800.00	5500 Day	182.20	74.84	Classroom	
9	Rm_028	9.70	8.72	2.59	95.42	84.58	1.00	1.80	1.89	3.00	15.10	15.83	2.62	850.00	5500 Day		44.70	52.85	Classroom
10	Rm_029	14.00	5.00	2.70	102.60	70.00	1.00	2.64	2.57	2.00	27.00	26.32	4.66	700.00	5500 Day		31.04	44.34	Cafeteria
11	Rm_030	9.80	5.00	2.30	68.08	49.00	1.00	1.97	2.89	3.00	17.40	25.56	6.57	850.00	5500 Day		21.45	43.78	Cafeteria
12																			
13																			
14																			
15																			

Standard Subject Data across all Experiments

	A	B	C	D	E	F	G
	Subject_ID	Name	age	gender	occupation		
1	Subj_A	Ameya Thopte		25 Female	Architect		
2	Subj_B	Sakshi Adwani		26 Female	Architect		
3	Subj_C	Rishabh Verma		28 Male	Architect		
4	Subj_D	Rishi Srinivasan		27 Male	Architect		
5	Subj_E	Flavia Soares		26 Female	Architect		
6	Subj_F	Srishti Kathuria		25 Female	Architect		

Subject Data

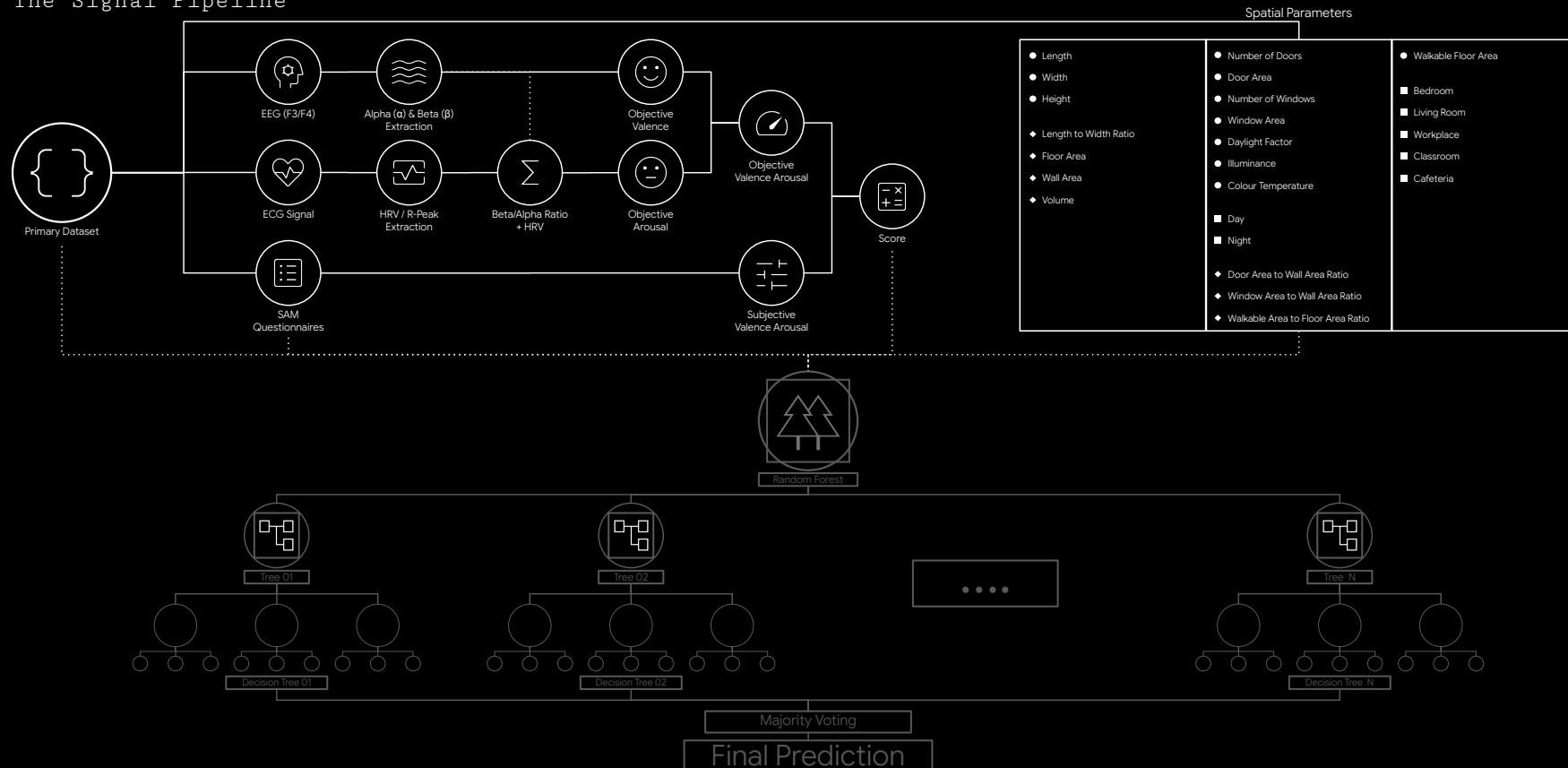
Individual Human Data of a Subject for a Room

	Counter	Channel1	Channel2	Channel3
1	218	0.079856	0.086934	-0.01941
2	219	0.080109	0.083123	-0.02011
3	220	0.079004	0.078728	-0.02272
4	221	0.077863	0.075595	-0.02762
5	222	0.077158	0.073181	-0.03332
6	223	0.075095	0.070451	-0.03789
7	224	0.073605	0.068594	-0.04041
8	225	0.074893	0.068944	-0.04074
9	226	0.076192	0.07028	-0.03791
10	227	0.074657	0.071974	-0.03177
11	228	0.073417	0.072841	-0.0243
12	229	0.073534	0.070526	-0.01872
13	230	0.070745	0.066928	-0.01657
14	231	0.065801	0.064923	-0.01575
15	232	0.062237	0.061817	-0.01506
16	233	0.0595583	0.058491	-0.01501
17	234	0.056343	0.057615	-0.01502
18	235	0.054703	0.054635	-0.01418
19	236	0.0544143	0.050542	-0.01127
20	237	0.052505	0.049999	-0.00541
21	238	0.051441	0.0523766	0.001839
22	239	0.0518917	0.054638	0.008278
23	240	0.051028	0.053099	0.012421
24	241	0.050269	0.049693	0.01372552
25	242	0.048732	0.0491431	0.01463546
26	243	0.045196	0.051181	0.01801192
27	244	0.04303	0.0510062	0.02396106
28	245	0.043653	0.045422	0.02939096
29	246	0.042654	0.038091	0.03124936
30	247	0.038199	0.0349533	0.031399
31	248	0.035859	0.035143	0.032521
32	249	0.037817	0.033466	0.0378290
33	250	0.0412659	0.033616	0.033984
34	251	0.0439	0.038286	0.034033
35	252	0.046028	0.0409443	0.034625
36	253	0.0466201	0.038988	0.034618

THE DATA PROCESSING

System Architecture

The Signal Pipeline



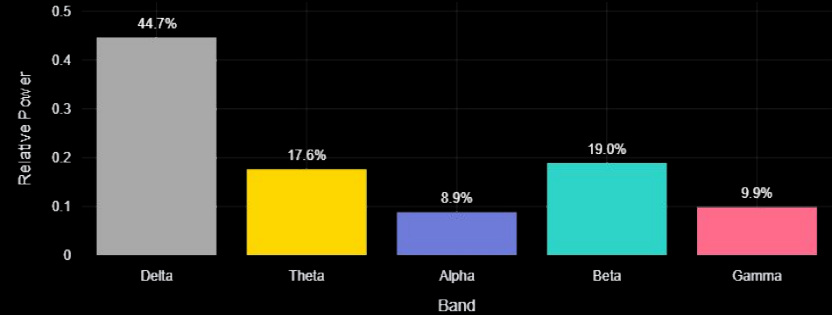
THE FINDINGS

Findings

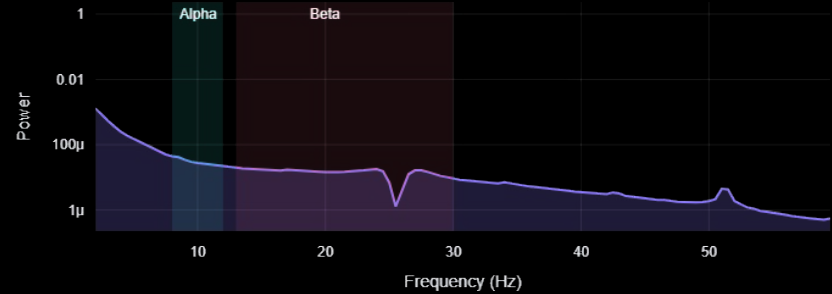
Raw Human Metrics and Filtering

Brain (EEG)

Average EEG Spectral Power

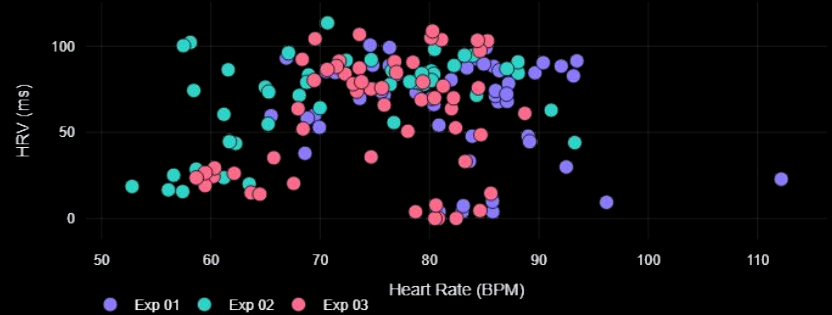


Brain Spectral Envelope

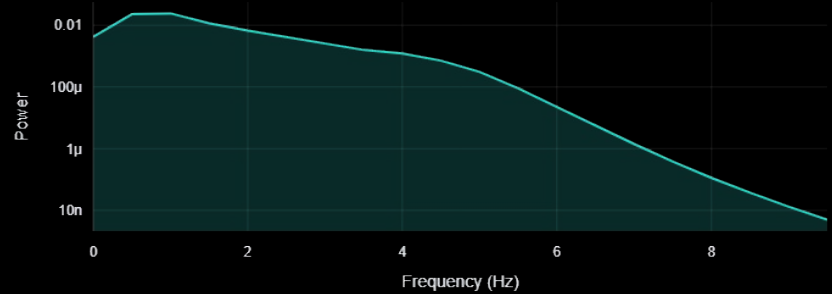


Heart (ECG)

Autonomic Regulation Map



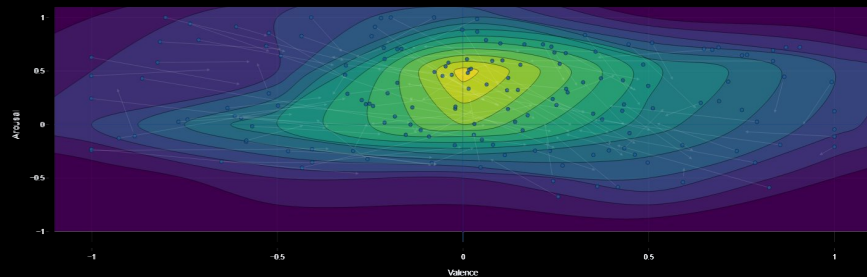
Cardiac Spectral Envelope



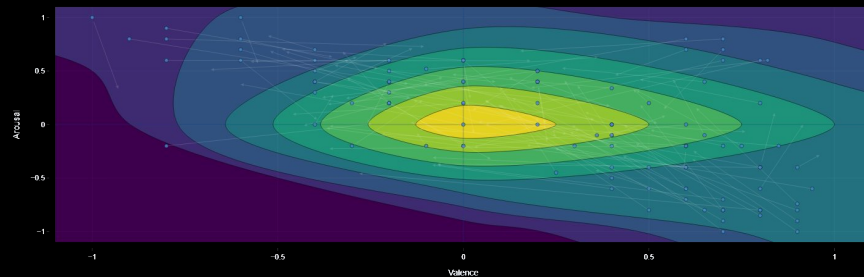
Findings

Valence-Arousal Density Field

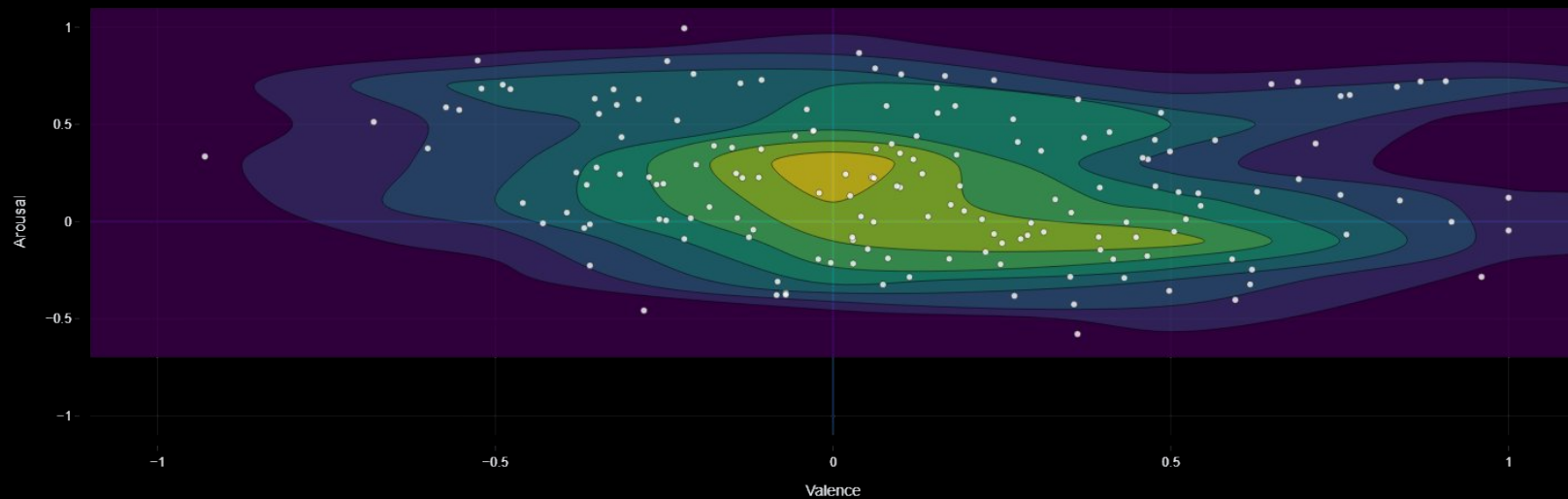
Objective Density Field



Subjective Density Field

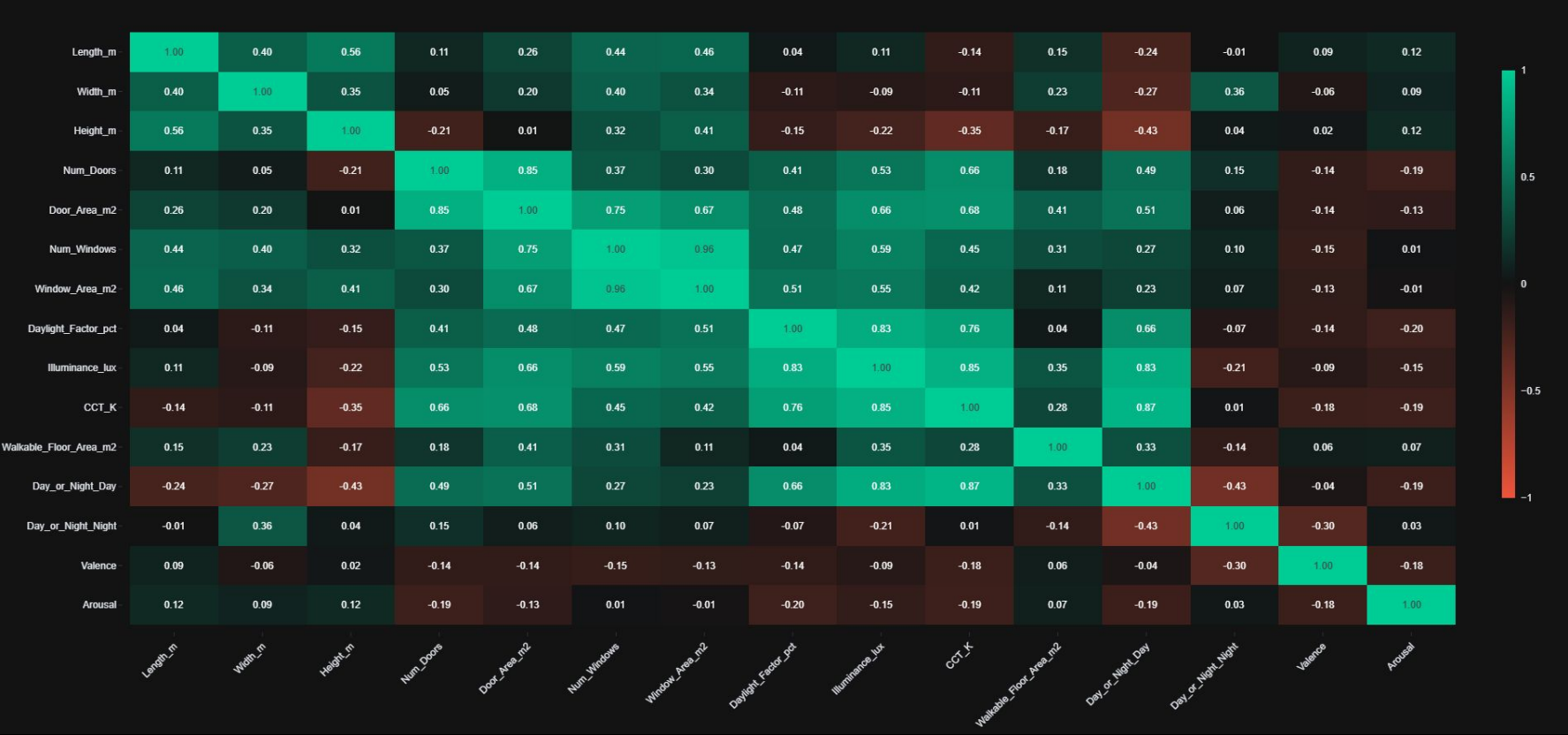


Target (Fused) Density Field



Findings

Feature Correlation Matrix

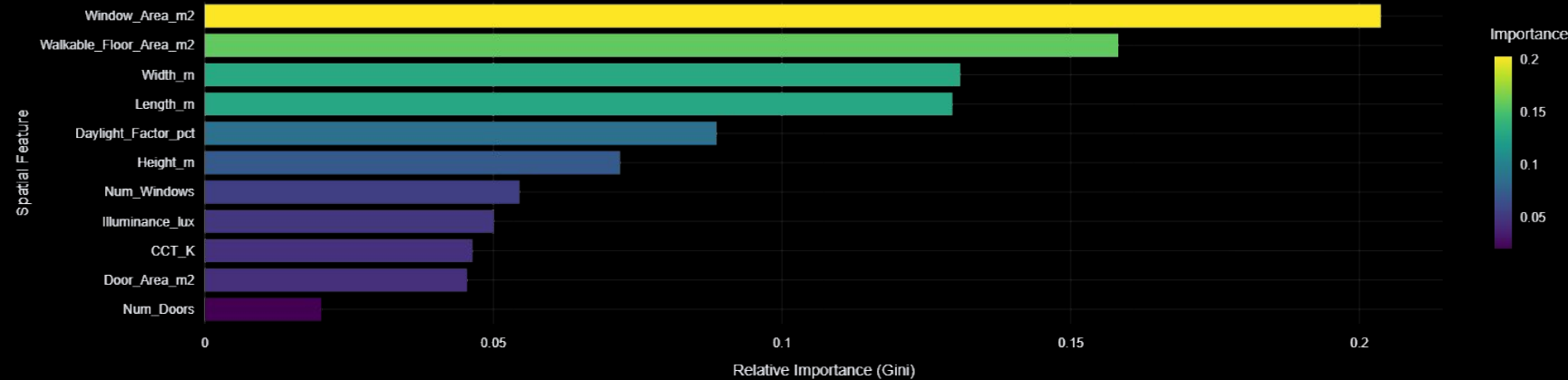


Findings

Spatial Insights

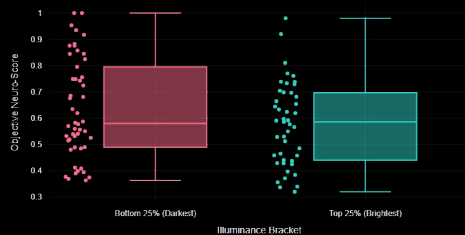
Non-Linear Feature Importance

Random Forest Feature Contributions



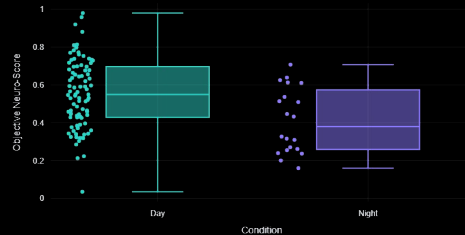
Day vs. Night Impact

Neuro-Score by Illuminance Extremes



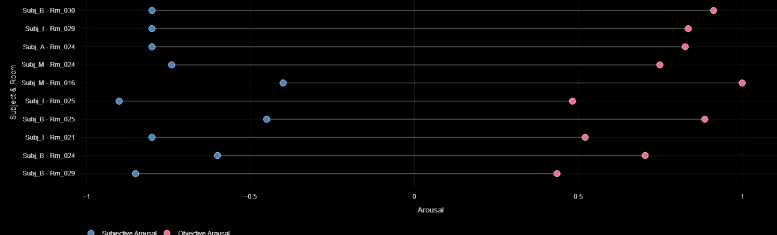
Illuminance Extremes

Objective Neuro-Score by Time of Day



Illuminance Extremes

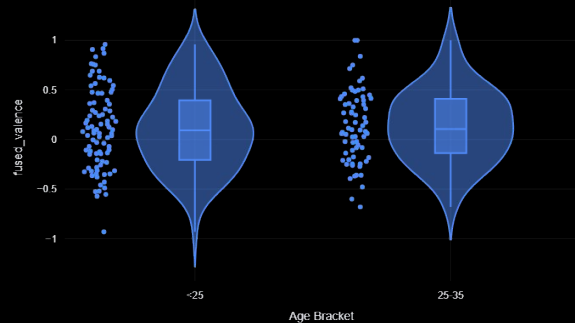
Spatial Triggers (Δ Arousal)



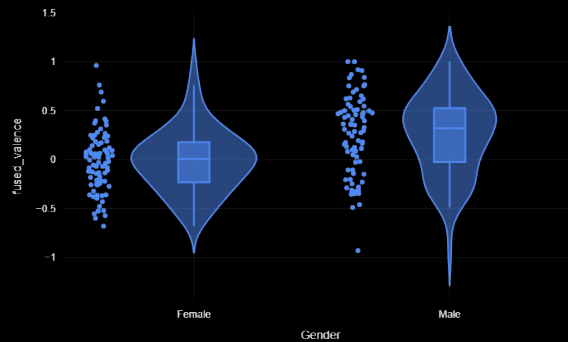
Findings

Subjective Data Insights

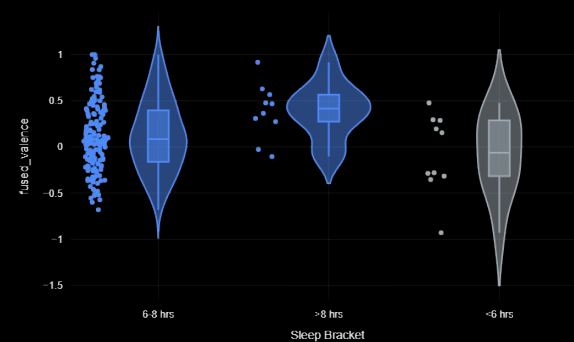
Target Valence by Age Bracket



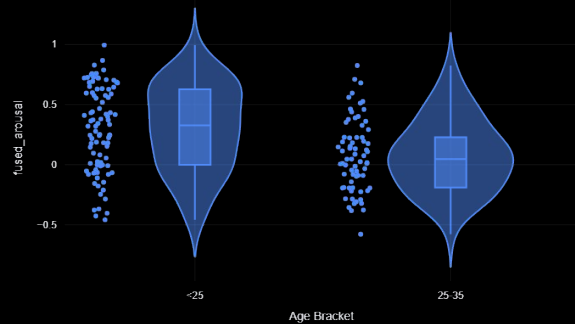
Target Valence by Gender



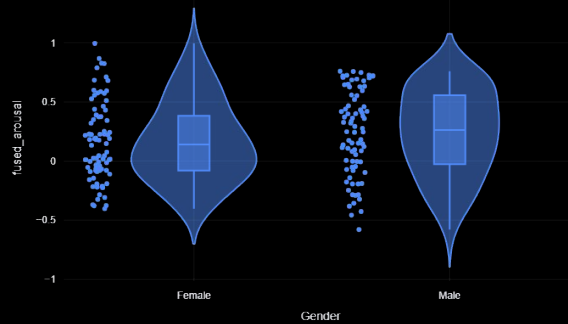
Target Valence by Sleep Bracket



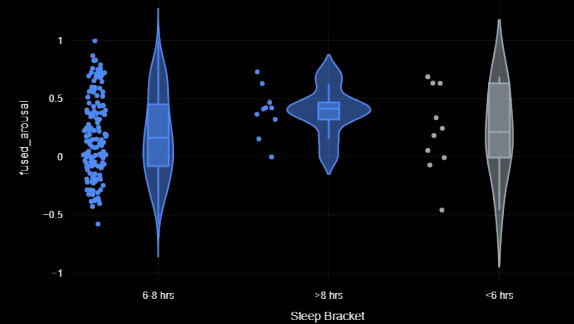
Target Arousal by Age Bracket



Target Arousal by Gender



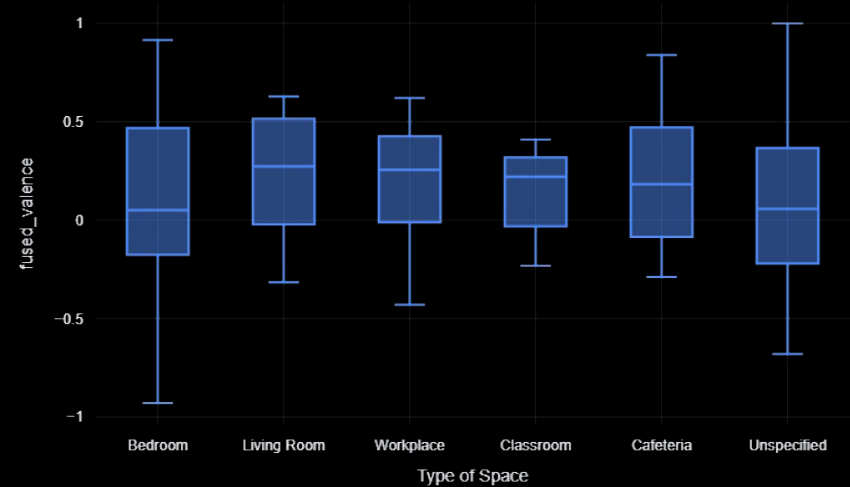
Target Arousal by Sleep Bracket



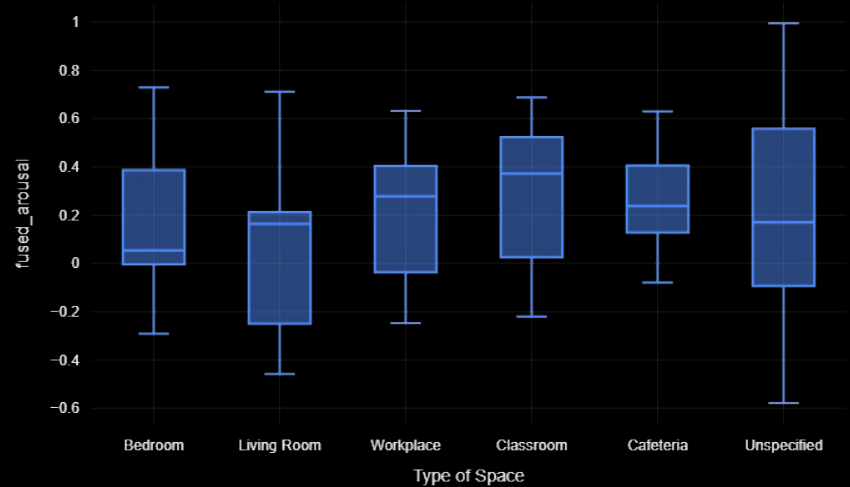
Findings

Environmental Impact

Target Valence by Space Type

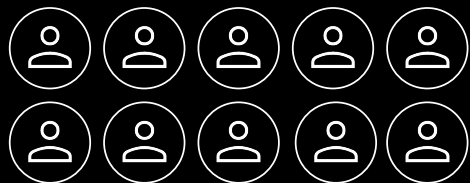


Target Arousal by Space Type



Limitations

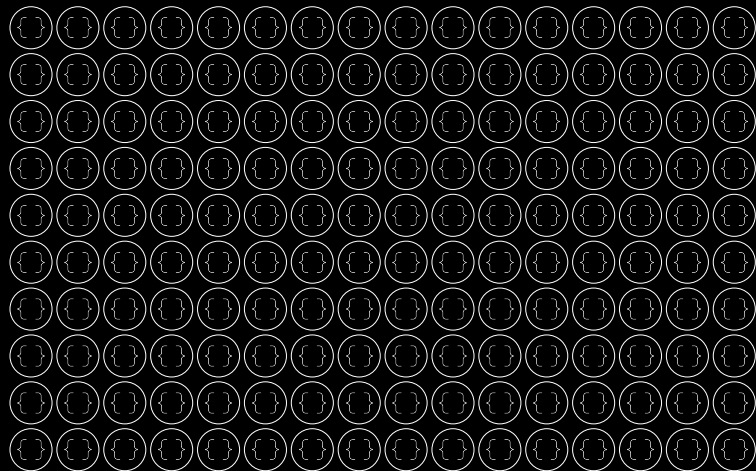
Data Scaling



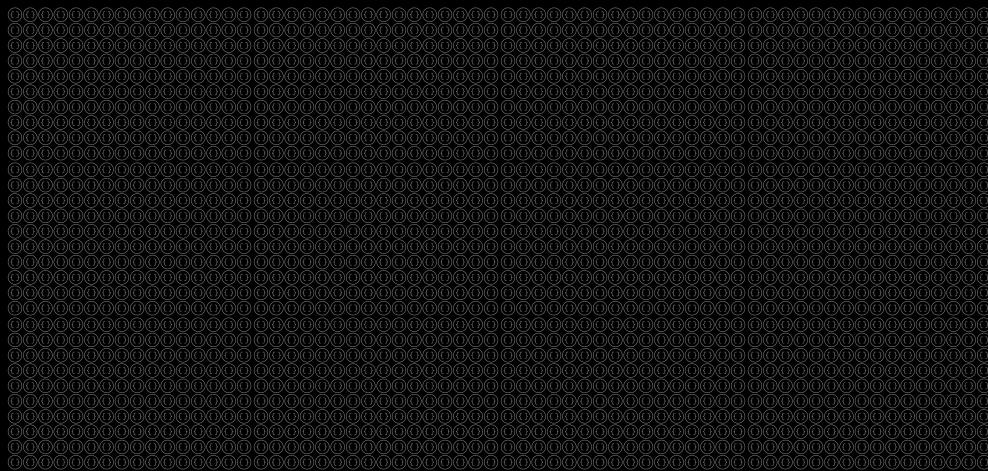
10 Subjects



10 X 40 = 400 Subjects (or more)



160 Data Samples

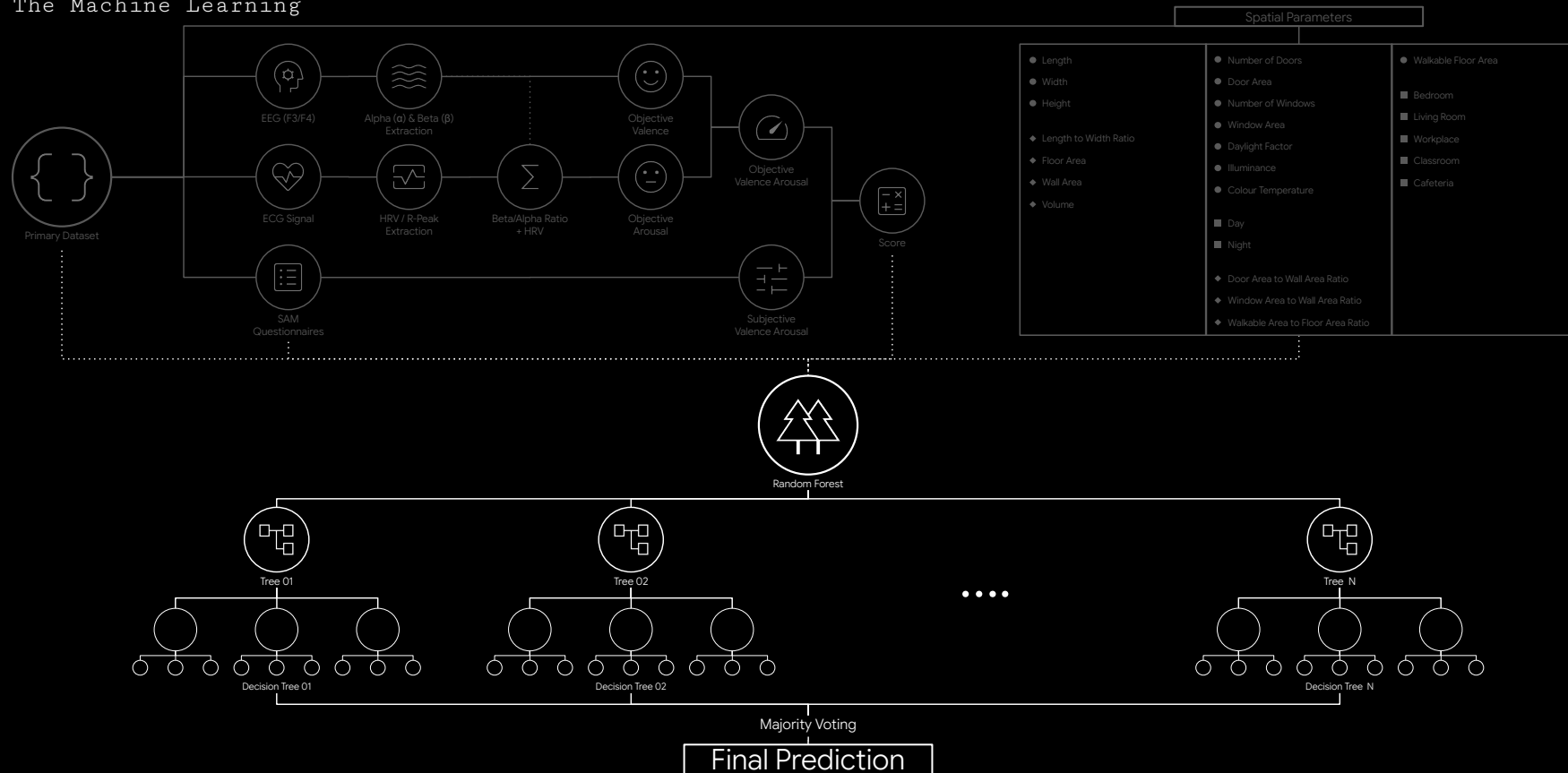


160 X 40 = 6400 Data Samples (or more)

THE MACHINE LEARNING

System Architecture

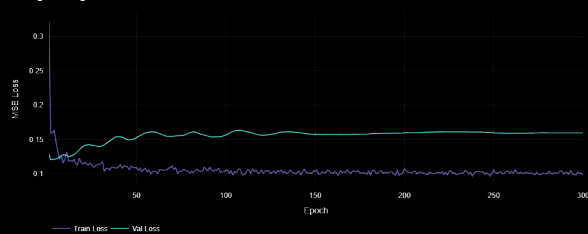
The Machine Learning



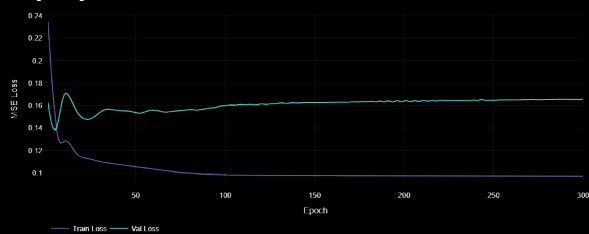
System Architecture

The Machine Learning

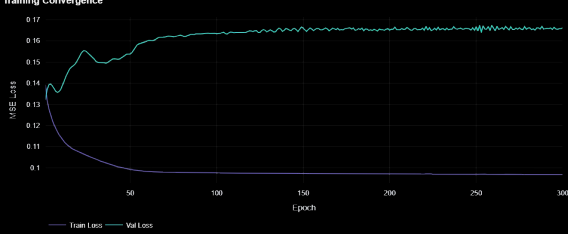
Training Convergence



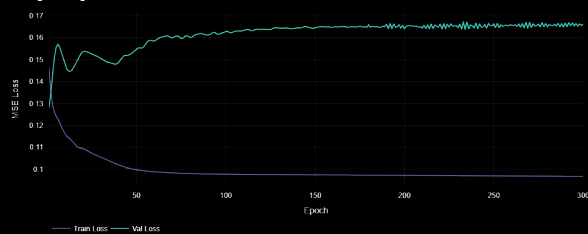
Training Convergence



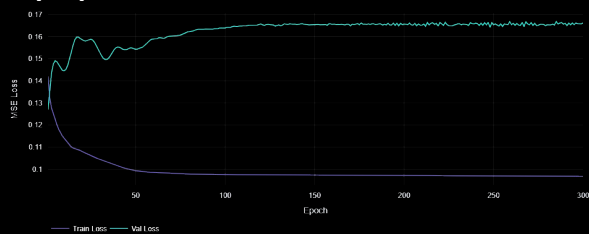
Training Convergence



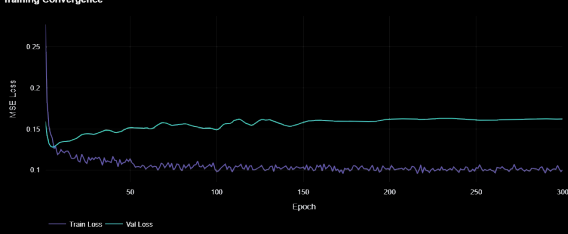
Training Convergence



Training Convergence



Training Convergence



THE PLATFORM

Interactive Design Studio

Shape room geometry and architectural parameters, inspect predicted emotional response, and use inverse optimization to co-design toward target affect.

Predicted Neuro-Score

0.53

Valence

0.26

Arousal

0.24

Confidence

76.1%

Atmospheric Label

NEUTRAL

Geometry

Length (m)

10.00

Width (m)

8.00

Height (m)

3.50

Daylight

Daylight Factor (%)

2.00

Illuminance (lux)

300.00

CCT (Kelvin)

4000.00

Openings

Number of Doors

1.00

Door Area (m²)

2.00

Number of Windows

2.00

Window Area (m²)

1.50

Space Details

Walkable Floor Area (m²)

24.00

Lighting Condition

☒ Day

☐ Night

Type of Space

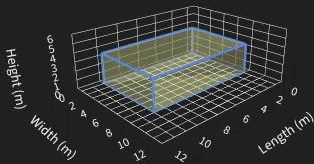
Unspecified

AI Co-Design

Target Neuro-Score

0.80

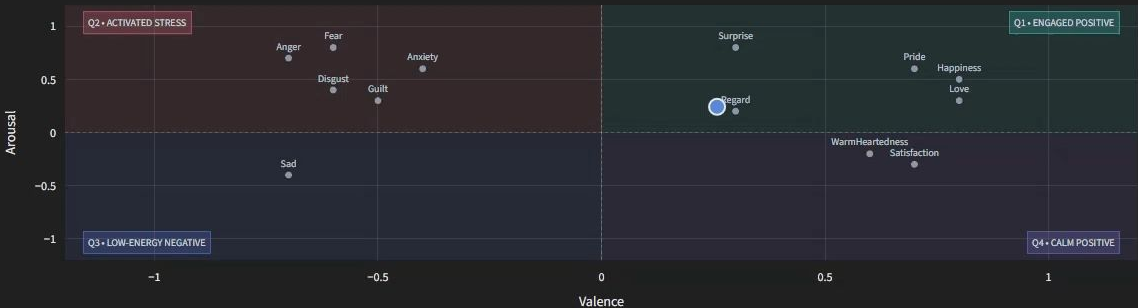
Auto-Design Room



Affective Map

Emotion Profile

Reference Alignment

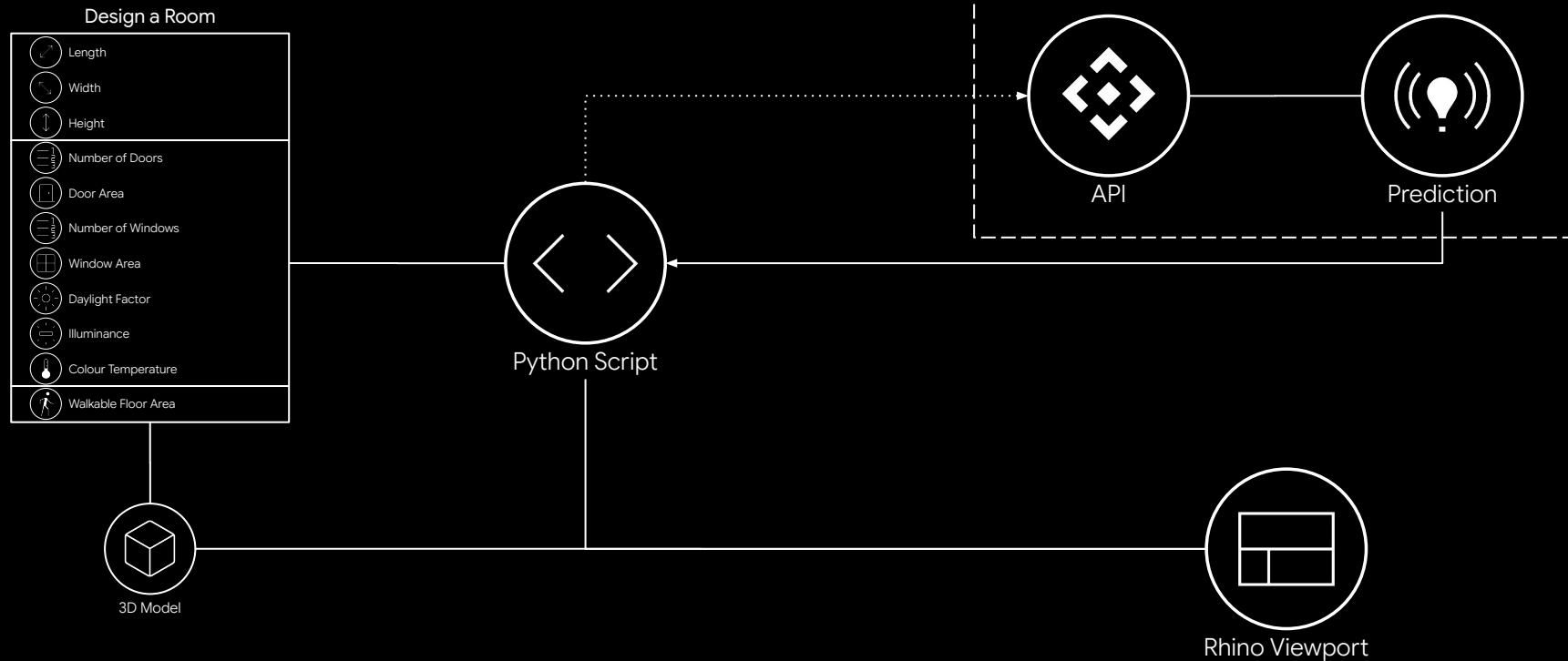


THE PLUGIN

Grasshopper Plugin

Methodology

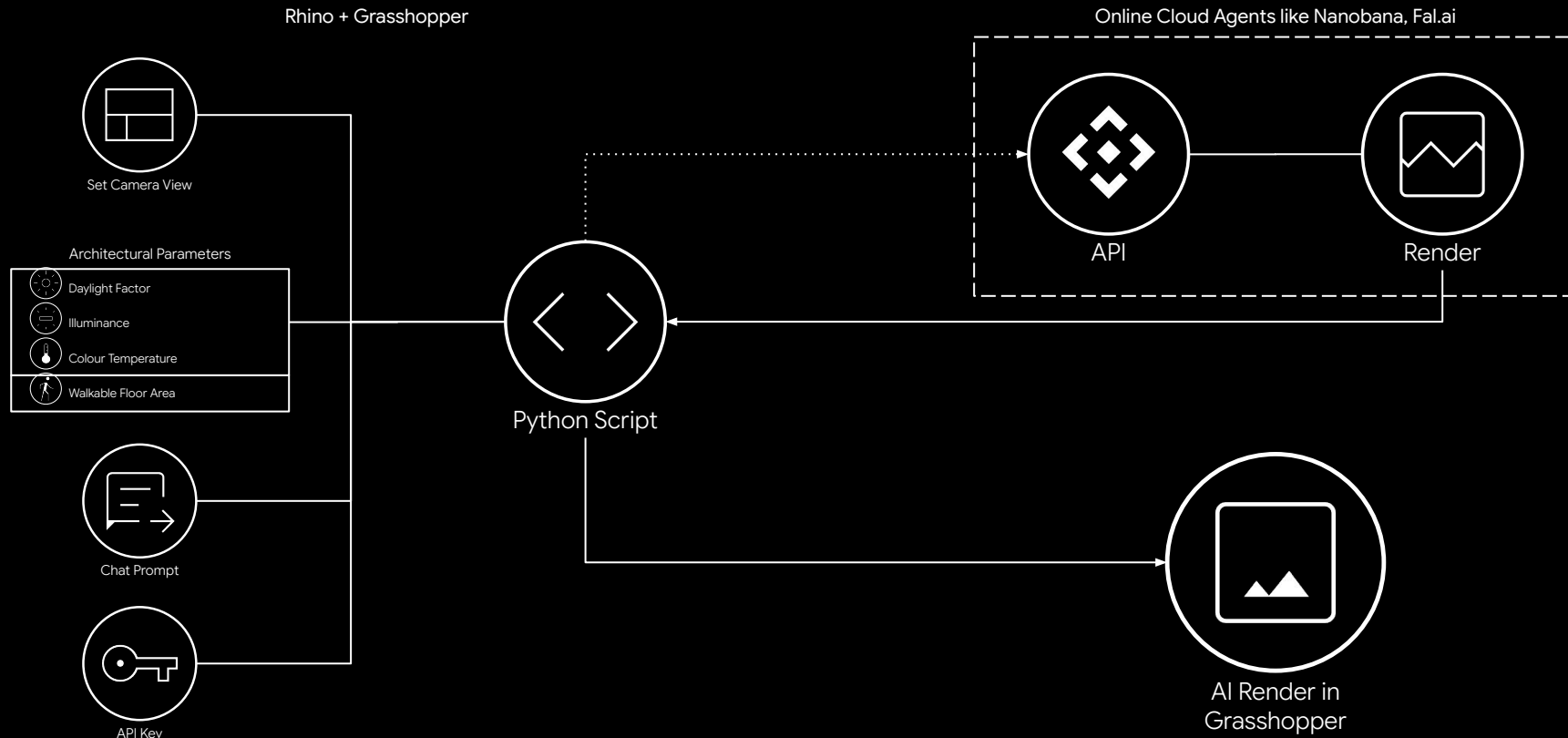
Rhino + Grasshopper



THE ATMOSPHERE

AI Rendering Pipeline

Methodology



Validation

By validating our hypothesis, this **milestone proves that emotional inference** can become the **new computational standard for design software**, evolving continuously with **scaled machine learning training.**



Predictive Atmospheres

Emotional Simulation for Architectural Design

is a thesis developed at
IAAC, Institute of Advanced Architecture of Catalonia for
Masters in Advanced Architecture
in 2025-26 by:

Student:
Manideep Mamidala

Thesis Advisors :
Firas Safieddine
Francisco Ramazzini Estivallet

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